

BlueBolt Innovation 2026

# QData

Conversational AI for Secure &  
Governed Enterprise Data Access

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AIA Data Vertical · Cognizant Technology Solutions



Production Ready



Enterprise Secure



Multi-Agent AI

# The Problem

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Data is locked behind SQL expertise.

## Business Users Blocked

No SQL expertise means every data request goes through a data engineer, no matter how simple.

## Engineers Bottlenecked

Data teams spend hours on repetitive ad-hoc queries instead of high-value engineering work.

## BI Tools Fall Short

Existing tools require training, have rigid interfaces, and lack intelligent multi-step reasoning.

*There is no lightweight, secure, and governable solution that lets users ask data questions in plain English and get accurate, audited answers — without exposing raw infrastructure or compromising data access policies.*

# The Solution

*"Type a question. Get the answer. No SQL. No waiting."*

**User** "Show me the top 10 customers by revenue this quarter"

**QData** Understands intent → Identifies schema → Executes SQL securely → Returns answer

"The top 10 customers by revenue this quarter are: Acme Corp (\$2.4M), GlobalTech (\$1.9M)..."

Plain English Input

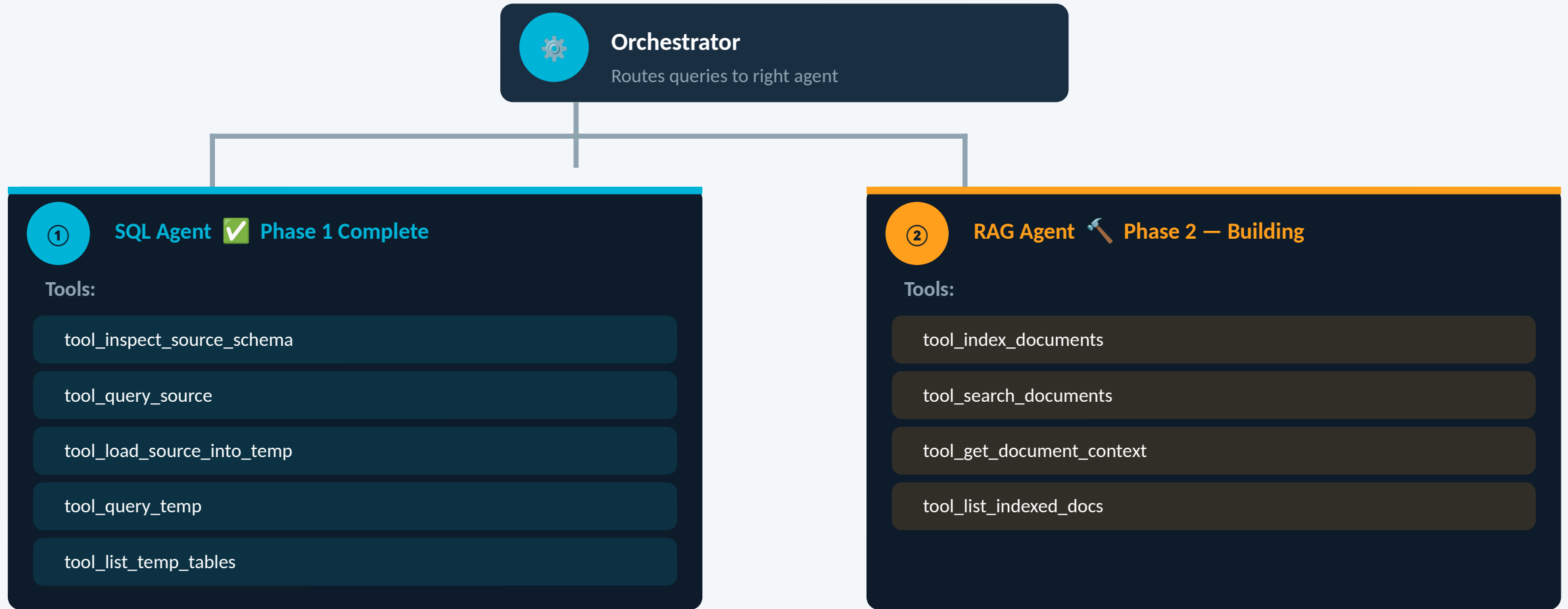
Auto SQL Generation

Governed Access

Full Audit Trail

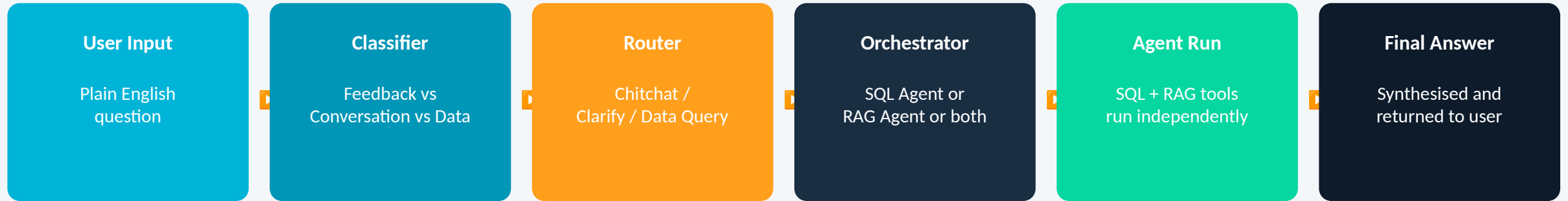
Built on a fully custom multi-agent architecture — SQL Agent + RAG Agent coordinated by an Orchestrator.

# Multi-Agent Architecture

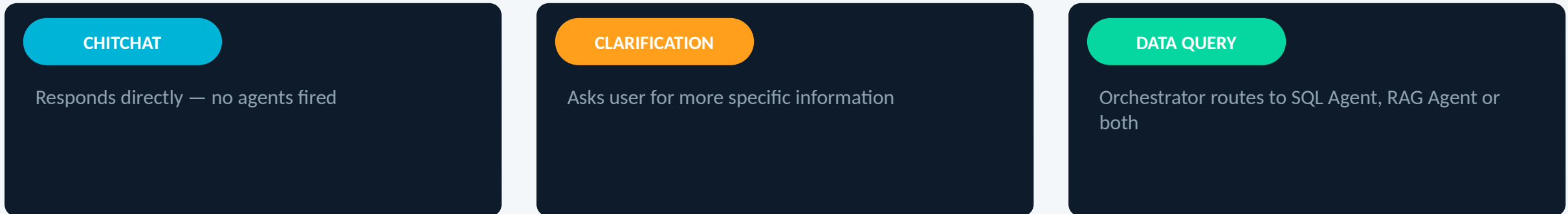


*Each agent has its own loop • own scratchpad • own tools • own LangSmith trace*

# How It Works



Router decides the path:



# Enterprise Security & Governance



## AST-Level SQL Validation

SQL parsed into syntax tree — INSERT, UPDATE, DELETE, DROP blocked at parse time, not by keywords.



## JWT Authentication

Every MCP request requires signed JWT. HS256 local secret for dev, AWS KMS RS256 for production.



## Redshift SSO

Queries run as the actual user. Redshift natively enforces table and row-level permissions.



## Full Audit Trail

Every run, every tool call logged in LangSmith with session ID, tools used, and termination reason.



## Session Isolation

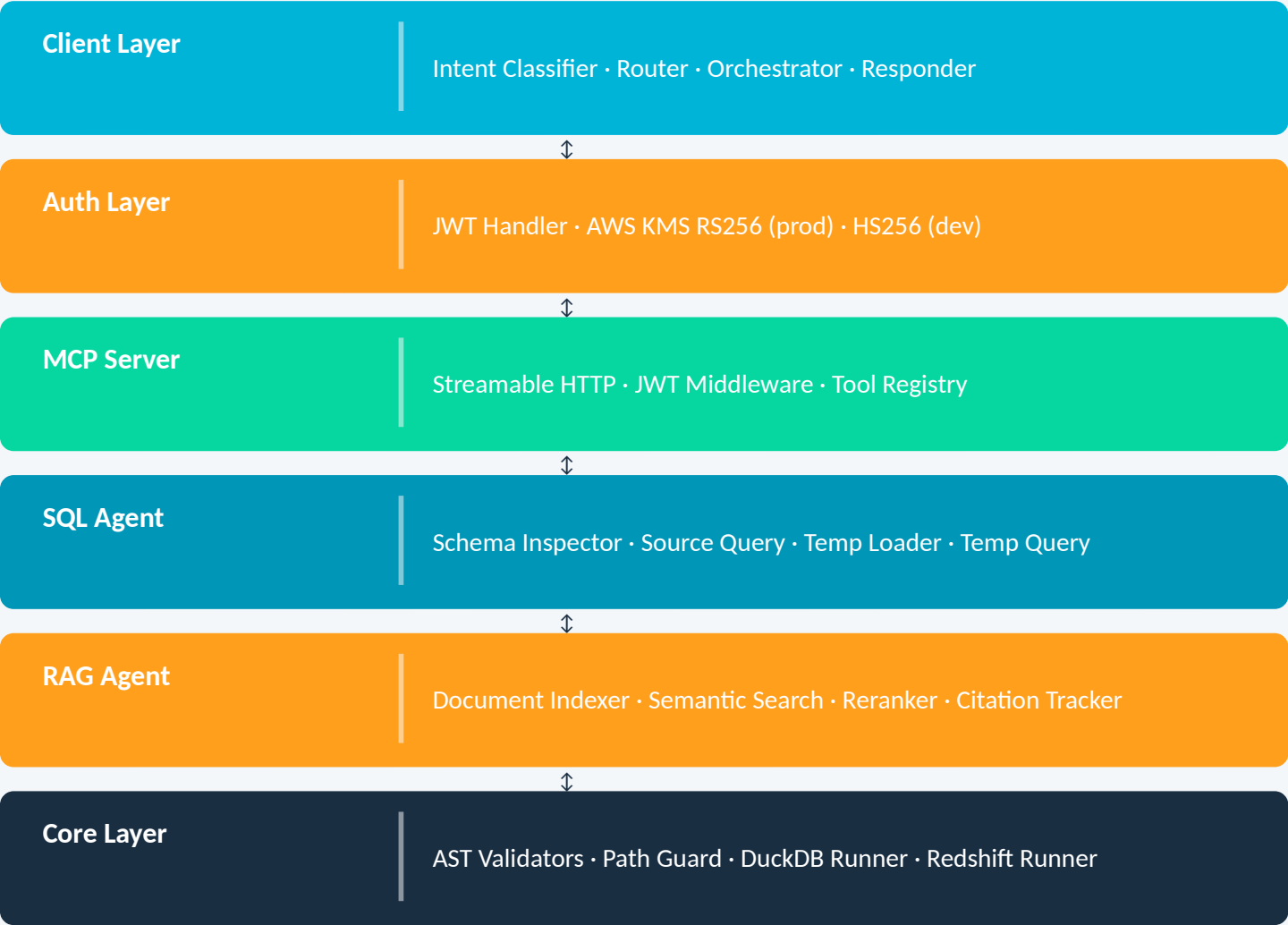
Each conversation gets its own isolated DuckDB sandbox. Temp data wiped automatically on session end.



## Structured Output

LLM forced to return Pydantic-validated schema. No prompt injection via malformed JSON possible.

# Technical Architecture



| Dev vs Production |            |              |
|-------------------|------------|--------------|
| Feature           | Dev        | Prod         |
| Data Source       | CSV+DuckDB | Redshift     |
| Auth Signing      | HS256      | KMS RS256    |
| Permissions       | Path Guard | Redshift SSO |
| SQL Agent         | ✓          | ✓            |
| RAG Agent         | ✓          | ✓            |
| AST Valid.        | ✓          | ✓            |
| LangSmith         | ✓          | ✓            |
| Switch            | ENV=dev    | ENV=prod     |

# Agent Intelligence

## Live Conversation

- You:** hi  
*CHITCHAT → responds directly*
- Bot:** Hello! How can I help you today?
- You:** show top 5 customers by revenue  
*DATA\_QUERY → SQL Agent*
- Bot:** The top 5 customers are: Acme (\$2.4M)...
- You:** summarise the Q3 report  
*DATA\_QUERY → RAG Agent*
- Bot:** Q3 report highlights: Revenue up 12%...
- You:** great thanks  
*POSITIVE\_FEEDBACK → logged*
- Bot:** Glad that helped!

## Agent Step-by-Step

Step 1

**tool\_inspect\_source\_schema()**

← Schema: sales → customers, orders

Step 2

**tool\_query\_source(sql)**

← SELECT ... FROM sales.customers...

Step 3

**tool\_search\_documents(query)**

← Q3 report: revenue up 12%...

Final

**Orchestrator synthesises both results**

← Combined answer returned to user

# Observability & Audit Trail

## LangSmith Trace — Every Run

Run: qdataagent-turn

chain

SQL Agent run

agent

tool\_inspect\_source\_schema

tool

tool\_query\_source

tool

RAG Agent run

agent

tool\_search\_documents

tool

Final Answer

output

1

### Per-Run Logging

Every turn logged with session ID, tools used, steps count, and termination reason.

2

### Child Run Tracing

Each tool call logged as a child run — full drill-down visibility per agent.

3

### User Feedback Scoring

Positive scores 1.0, negative 0.0 — tied to run\_id for complete traceability.

4

### Token Usage Tracking

Input, output, total token counts per run for cost monitoring and optimisation.

# Business Impact



## Seconds

Ad-hoc request  
resolution vs days



## Zero

SQL expertise  
needed by users



## 100%

Queries audited  
and traceable



## Any

Client warehouse  
deployable instantly

### Reusability Across AIA Data Vertical

QData requires minimal configuration change per client — any organisation running Amazon Redshift, Snowflake, or a structured data warehouse can deploy this multi-agent system and immediately enable natural language data access for their teams.

# Current Status & Roadmap

## Phase 1 — SQL Agent Complete

- ✓ Fully working terminal-based agent
- ✓ Amazon Redshift + CSV connectivity
- ✓ Multi-schema dot notation queries
- ✓ AST-level SQL mutation prevention
- ✓ JWT auth (HS256 dev / KMS RS256 prod)
- ✓ LangSmith audit trail — every run traced
- ✓ Production-grade intent router
- ✓ Structured output — zero hallucinated calls
- ✓ Session isolation and temp sandbox
- ✓ MCP server — tools as HTTP service

## Roadmap — Next Phases

### Phase 2

### RAG Agent

PDF ingestion, semantic chunking, hybrid vector+BM25 search, reranking, citation tracking. Connects as second agent to orchestrator.

### Phase 3

### Web UI

React chat interface for end users. File upload for CSV and PDFs. REST API via FastAPI app server.

### Phase 4

### Docker + AWS ECS

Containerised deployment. docker-compose for local. Scalable cloud deployment on ECS.

### Phase 5

### Multi-Agent Orchestration

Parallel SQL + RAG execution. Cross-domain synthesis. Complex multi-hop analytical workflows.

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Submitted under BlueBolt Innovation Program

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Agents  
(SQL + RAG)

6+

Security  
Layers

100%

Production  
Ready